



MEREWETHER
HIGH SCHOOL

2018

HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK 4

Mathematics Extension 1

**General
Instructions**

- Reading time – 5 minutes
- Working time – 2 hours.
- Write using black pen.
- NESA approved calculators may be used.
- A reference sheet is provided at the back of this paper.
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations.
- Submit ALL work at the end of the examination.
- Diagrams are NOT drawn to scale.

**Total marks:
70****Section I – 10 marks** (pages 2 – 5)

- Attempt Questions 1 – 10.
- Allow about 15 minutes for this section.

Section II – 60 marks (pages 6 – 12)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section.

Section I

10 marks

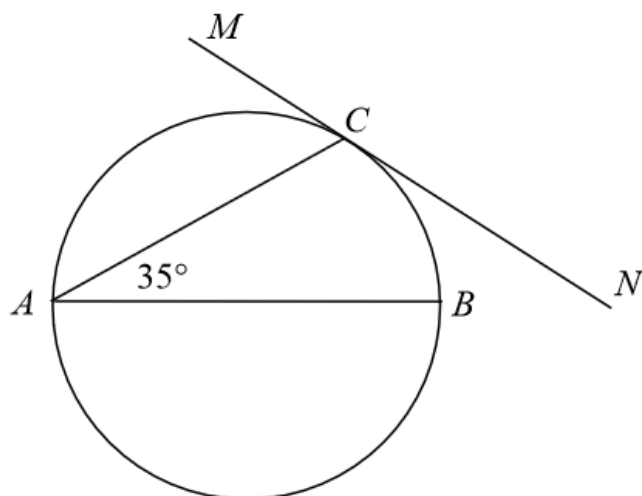
Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

- 1 The equation $2x^3 + x^2 - 13x + 6 = 0$ has roots α , $\frac{1}{\alpha}$ and β . What is the value of β ?
- A. 3
B. -3
C. 2
D. -6
- 2 Which of the following is a simplification of $4\log_e \sqrt{e^x}$?
- A. $4\sqrt{x}$
B. $\frac{1}{2}x$
C. $2x$
D. x^2
- 3 Which of the following is an expression for $\frac{dy}{dx}$ if $x = \frac{1}{2}at$ and $y = at^2$?
- A. t
B. $2t$
C. $2at$
D. $4t$

- 4 In the triangle, AB is the diameter of the circle and MCN is the tangent to the circle at C . $\angle CAB = 35^\circ$. What is the size of $\angle MCA$?



- A. 35°
B. 45°
C. 55°
D. 65°
- 5 Which of the following is an expression for $\int \sin^2 2x dx$?
- A. $\frac{1}{2}x - \frac{1}{8}\sin 4x + C$
B. $\frac{1}{2}x - \frac{1}{4}\sin 4x + C$
C. $\frac{1}{2}x + \frac{1}{8}\sin 4x + C$
D. $\frac{1}{2}x + \frac{1}{4}\sin 4x + C$

- 6 What is the value of $\lim_{x \rightarrow 0} \left(\frac{\sin \frac{1}{3}x}{2x} \right)$?
- A. $\frac{1}{6}$
- B. $\frac{2}{3}$
- C. $\frac{3}{2}$
- D. 6
- 7 Which of the following is the range of the function $y = 2 \sin^{-1} x + \frac{\pi}{2}$?
- A. $-\pi \leq y \leq \pi$
- B. $-\frac{\pi}{2} \leq y \leq \frac{3\pi}{2}$
- C. $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
- D. $-\pi \leq y \leq \frac{3\pi}{2}$
- 8 Which of the following is an expression for $\frac{d}{dx} \sin^{-1}(2x-1)$?
- A. $\frac{-1}{\sqrt{x(x-1)}}$
- B. $\frac{-1}{2\sqrt{x(x-1)}}$
- C. $\frac{1}{2\sqrt{x(x-1)}}$
- D. $\frac{1}{\sqrt{x(1-x)}}$

- 9 If $\frac{dN}{dt} = 0.1(N - 100)$ and $N = 300$ when $t = 0$, which of the following is an expression for N ?
- A. $200 + 100e^{0.1t}$
B. $300 + 100e^{0.1t}$
C. $100 + 200e^{0.1t}$
D. $100 + 300e^{0.1t}$
- 10 A particle is performing simple harmonic motion in a straight line. At time t seconds it has displacement x metres from a fixed point O on the line where x is given by $x = 4\sin^2 t - 1$. Where is the centre of motion?
- A. $x = -1$
B. $x = 0$
C. $x = 1$
D. $x = 2$

End of Section I

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Solve $\frac{4x-1}{x+2} \geq 1$ 2

(b) $A(-3,1)$ and $B(1,-2)$ are two points. Find the coordinates of point P that divides the interval AB externally in the ratio $3:1$. 2

(c) Find the number of ways in which the letters of the word CIRCLE can be arranged in a row so that the two vowels are in the two end positions. 2

(d) $P(2at, at^2)$ is a point on the parabola $x^2 = 4ay$ with focus $F(0, a)$

(i) Show that the normal to the parabola at P has gradient $-\frac{1}{t}$. 1

(ii) If θ is the acute angle between the normal to the parabola and the line PF show that $\tan \theta = |t|$. 2

Question 11 continues on page 7

(e) The equation $x^3 - 6x^2 + 4x + 2 = 0$ has roots α , β and γ . Find the value of

(i) $\frac{\alpha}{2} + \frac{\beta}{2} + \frac{\gamma}{2}$ **1**

(ii) $\frac{2}{\alpha} + \frac{2}{\beta} + \frac{2}{\gamma}$ **2**

(f) Use the substitution $u = x + 2$ to find $\int \frac{3x+5}{\sqrt{x+2}} dx$. **3**

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) Find $\frac{d}{dx}(e^x \tan^{-1} x)$. 2

(b) Use the substitution $t = \tan \frac{x}{2}$ to show that $\frac{1 + \cos x + \sin x}{1 - \cos x + \sin x} = \cot \frac{x}{2}$. 3

(c) Use the method of Mathematical Induction to show that $5^n - 4n - 1$ is divisible by 4 for all positive integers $n \geq 2$. 3

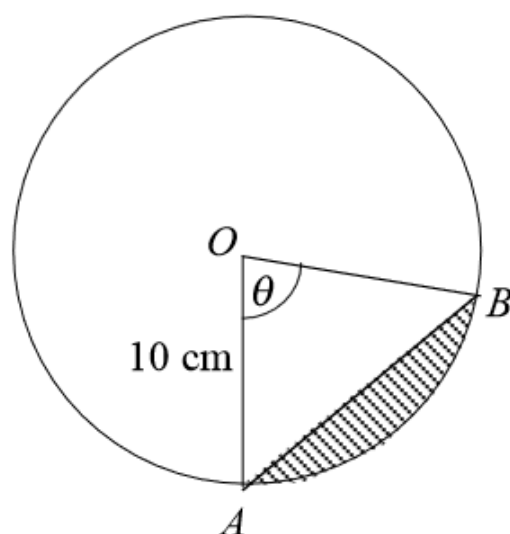
(d) Consider the function $f(x) = 2 \cos^{-1}(x - 1)$ where $1 \leq x \leq 2$.

(i) Sketch the curve $y = f(x)$, showing clearly the coordinates of the endpoints. 2

(ii) Find the equation of the inverse function $f^{-1}(x)$ and state its domain. 2

Question 12 continues on page 9

(e)



A circle has centre O and radius 10 cm. OA is a fixed radius of the circle. OB is a variable radius which moves so that $\angle AOB = \theta$ is increasing at a constant rate of 0.01 radians per second. The minor segment of the circle cut off by the chord AB has shaded area $S \text{ cm}^2$.

Find the rate at which S is increasing when $\theta = \frac{\pi}{3}$.

3

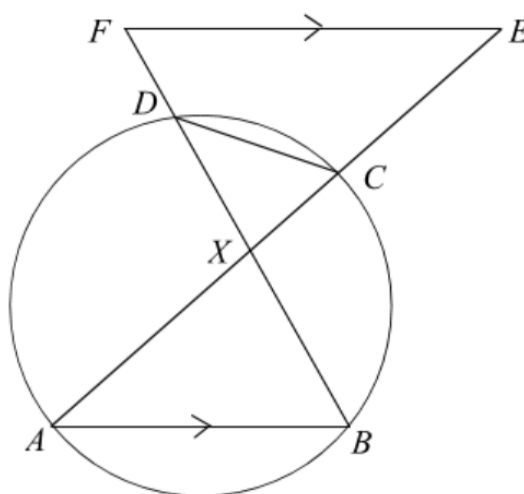
End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) Consider the equation $x^3 + 2x - 7 = 0$

- (i) Show that the equation has a root α such that $1 < \alpha < 2$. 2
- (ii) Show that α is the only real root of the equation. 1
- (iii) Use one application of Newton's Method with an initial approximation $\alpha = 1.5$ to find the next approximation for α correct to 1 decimal place. 2

(b) 3

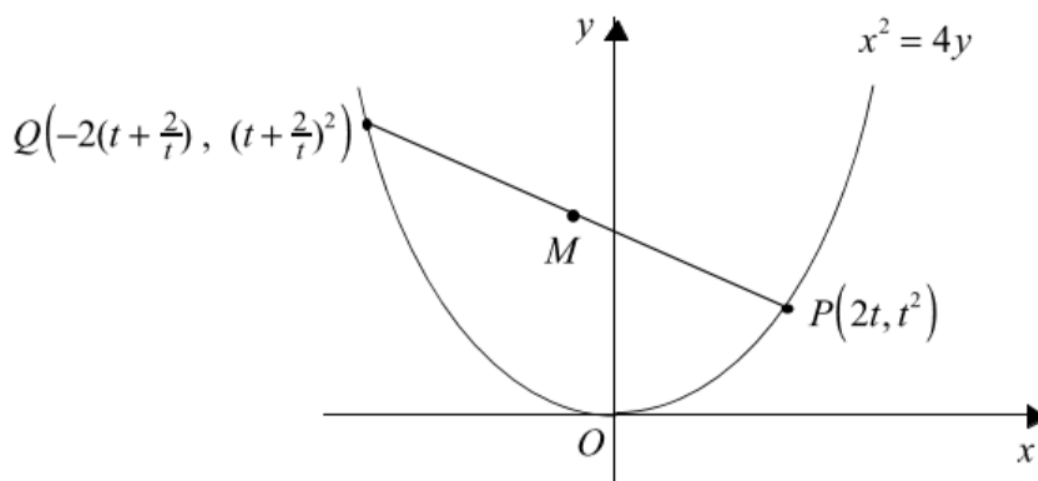


AC and BD are two chords of a circle which intersect at point X inside the circle.
 E is a point on AC produced and F is a point of BD produced such that $FE \parallel AB$.
 Show that $DCEF$ is a cyclic quadrilateral.

(c) Find the exact value of $\int_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}} \frac{1}{\sqrt{4-x^2}} dx$. 3

Question 13 continues on page 11

(d)



$P(2t, t^2)$ and $Q(-2(t + \frac{2}{t}), (t + \frac{2}{t})^2)$ are two points on the parabola $x^2 = 4y$.

M is the midpoint of PQ

- (i) By considering the gradient of QP , show that QP is normal to the parabola at P . 2
- (ii) Find the equation of the locus at M as P moves on the parabola. 2

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a) If $\sin^{-1} x - \cos^{-1} x = \frac{\pi}{3}$, show that $\sin^{-1} x = \frac{5\pi}{12}$. 2

(b) A particle moves along the x – axis according to the equation $x = \sin 2t - \sqrt{3} \cos 2t$, where x is in metres and t is in seconds.

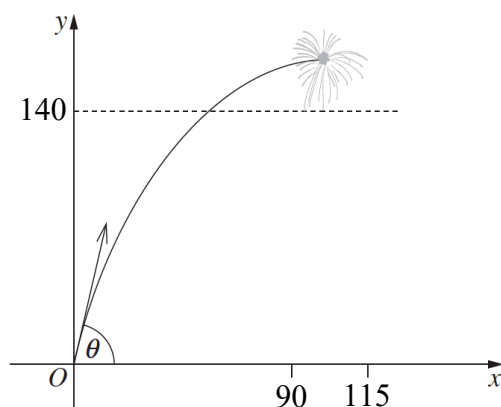
(i) Express x in the form $R \sin(2t - \alpha)$, where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. 2

(ii) Show that the particle is moving in simple harmonic motion. 2

(iii) State the period of the particle's motion. 1

(iv) When is the first time that the particle is 2 metres right of the origin? 2

(c) A firework is fired from O , on level ground, with velocity 60 metres per second at an inclination θ . The equations of motion of the firework are $x = 60t \cos \theta$ and $y = 60t \sin \theta - 5t^2$ (Do NOT prove this). The firework explodes when it reaches its maximum height.



**NOT TO
SCALE**

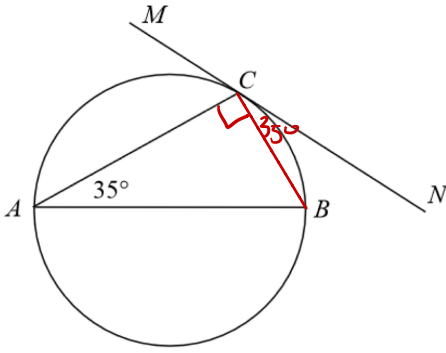
(i) Show that the firework explodes at a height of $180 \sin^2 \theta$ metres. 2

(ii) Show that the firework explodes at a horizontal distance of $180 \sin 2\theta$. 1

(iii) For best viewing, the firework must explode at a horizontal distance between 90 metres and 115 metres from O , and at least 140 metres above the ground. For what values of θ will this occur? 3

End of paper

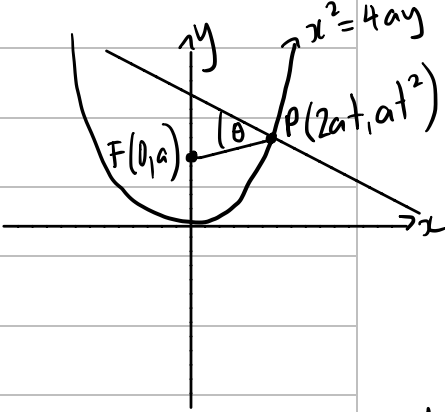
Question	Suggested solution	Marking guidelines	Markers Notes
1.	$d \times \frac{1}{a} \times \beta = -\frac{d}{a}$ $\beta = \frac{-6}{2}$ $= -3$	B	
2.	$4 \log_e \sqrt{e^x} = 4 \log_e e^{\frac{x}{2}}$ $= \cancel{2} \cancel{4} \left(\frac{x}{\cancel{2}} \right) \log_e e$ $= 2x$	C	
3.	$x = \frac{1}{2}at \quad y = at^2$ $\frac{dx}{dt} = \frac{a}{2} \quad \frac{dy}{dt} = 2at$ $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$ $= \cancel{2}at \times \frac{2}{\cancel{a}}$ $= 4t$	D	Don't sub $a = \frac{2x}{t}$ into y for $y = 2xt$ to find $\frac{dy}{dx}$. This is not differentiable as 3 variables are involved.

Question	Suggested solution	Marking guidelines	Markers Notes
4.	 NOT TO SCALE $\angle ACB = 90^\circ$ ($\angle$ in semi circle) $\angle NCB = 35^\circ$ ($\angle$ in alternate segment) $\therefore \angle MCA = 180 - (90 + 35)$ ($\angle$'s on straight line) $= 55^\circ$	C	
5.	$\int \sin^2 2x \, dx = \int \frac{1}{2} (1 - \cos 4x) \, dx$ $= \int \left(\frac{1}{2} - \frac{1}{2} \cos 4x \right) \, dx$ $= \frac{1}{2} x - \frac{1}{8} \sin 4x + C$	A	
6.	$\lim_{x \rightarrow 0} \frac{\sin \frac{1}{3}x}{2x} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{\sin \frac{1}{3}x}{x}$ $= \frac{(\frac{1}{3})}{2} \lim_{x \rightarrow 0} \frac{\sin \frac{1}{3}x}{\frac{1}{3}x}$ $= \frac{1}{6} \times 1$ $= \frac{1}{6}$	A	

Question	Suggested solution	Marking guidelines	Markers Notes
7.	$y = 2 \sin^{-1} x + \frac{\pi}{2}$ range of $y = \sin^{-1} x$: $-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}$ $-\pi \leq 2 \sin^{-1} x \leq \pi$ $-\frac{\pi}{2} \leq 2 \sin^{-1} x + \frac{\pi}{2} \leq \frac{3\pi}{2}$ $\therefore$ range of $y = 2 \sin^{-1} x + \frac{\pi}{2}$: $-\frac{\pi}{2} \leq y \leq \frac{3\pi}{2}$	B	
8.	$\frac{d}{dx} \sin^{-1}(2x-1) = \frac{1}{\sqrt{1-(2x-1)^2}} \times 2$ $= \frac{2}{\sqrt{4x-4x^2}}$ $= \frac{2}{\sqrt{4x(1-x)}}$ $= \frac{\cancel{2}}{\cancel{2} \sqrt{x(1-x)}}$ $= \frac{1}{\sqrt{x(1-x)}}$	D	

Question	Suggested solution	Marking guidelines	Markers Notes
9	$\frac{dN}{dt} = 0.1(N-100)$ $\therefore N = 100 + Ae^{0.1t} \text{ for some constant } A$. When $N=300$ and $t=0$ $300 = 100 + Ae^{0.1(0)}$ $200 = A$ $\therefore N = 100 + 200e^{0.1t}$	C	
10.	$x = 4\sin^2 t - 1$ $= 2(1 - \cos 2t) - 1$ $= 1 - 2\cos 2t$ $\therefore$ centre is at $x=1$ B C D C A A B D C C	C	

Question	Suggested solution	Marking guidelines	Markers Notes
11.(a)	$\frac{4x-1}{x+2} \geq 1$ $x+2 \neq 0$ $x \neq -2$ Solve $\frac{4x-1}{x+2} = 1$ $4x-1 = x+2$ $3x = 3$ $x = 1$ Test $x = -3$ $\frac{4(-3)-1}{-3+2} = 13 \geq 1$ ✓ $x = 0$ $\frac{4(0)-1}{0+2} = -\frac{1}{2} \leq 1$ ✗ $x = 2$ $\frac{4(2)-1}{2+2} = \frac{7}{4} \geq 1$ ✓ $\therefore x < -2, x \geq 1.$	1 1	Must show substitution for test
(b)	$A(-3, 1)$ $B(1, -2)$ externally $m:n$ $x = \frac{mx_2 + nx_1}{m+n}$ $y = \frac{my_2 + ny_1}{m+n}$ $= \frac{3(1) - 1(-3)}{3-1}$ $= \frac{6}{2}$ $= 3$ $\therefore P(3, -\frac{7}{2})$	$\frac{3(-2) - 1(1)}{3-1}$ $= \frac{-7}{2}$ ① ①	

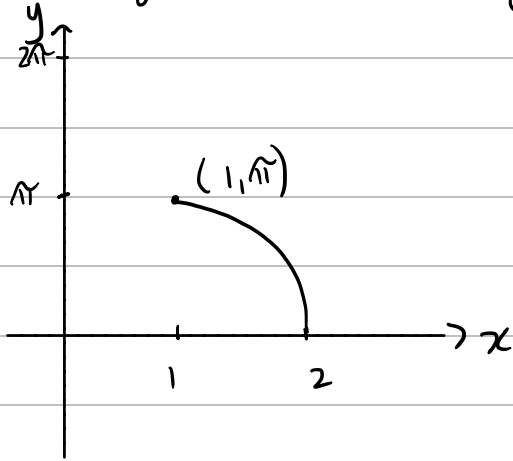
Question	Suggested solution	Marking guidelines	Markers Notes
11(c)	6 letters, 2 c's 4 consonants 2 vowels <u>v</u> <u>c</u> <u>c</u> <u>c</u> <u>c</u> <u>v</u> $\therefore$ Total arrangements = $\frac{2! \times 4!}{2!}$ $= 4!$ $= 24$	1 1	* Forgetting to divide by 2! for the 2 c's.
(d) i)	$x^2 = 4ay$ $y = \frac{x^2}{4a}$ $\frac{dy}{dx} = \frac{2x}{4a}$ $= \frac{x}{2a}$ $\therefore$ gradient of tangent at P, $\frac{dy}{dx} = \frac{2at}{2a}$ $= t$ $\therefore$ gradient of normal = $-\frac{1}{t}$	1	
ii)	$m_{PF} = \frac{at^2 - a}{2at - 0}$ $= \frac{a(t^2 - 1)}{2at}$ $= \frac{t^2 - 1}{2t}$	 1	must use differentiation for show that

Question	Suggested solution	Marking guidelines	Markers Notes
	$\tan \theta = \left \frac{\frac{t^2-1}{2t} - \left(-\frac{1}{t}\right)}{1 + \frac{1}{2t}(t^2-1)\left(-\frac{1}{t}\right)} \right $ $\left \frac{\frac{1}{2t}[(t^2-1) - (-2)]}{1 + \frac{1}{2t}(t^2-1)\left(-\frac{1}{t}\right)} \right $ $= \left \frac{\frac{1}{2t}(t^2+1)}{1 - \frac{1}{2t^2}(t^2-1)} \right $ $= \left \frac{\cancel{\frac{1}{2t}}(t^2+1)}{\cancel{\frac{1}{2t}}(2t^2-1)(\cancel{t^2-1})} \right $ $= \left \frac{t(t^2+1)}{2t^2 - t^2 + 1} \right $ $= \left \frac{t(\cancel{t^2+1})}{\cancel{t^2+1}} \right $ $= t $		* struggled with algebra required
(e)	$x^3 - 6x^2 + 4x + 2 = 0$	1	
(i)	$\frac{a}{2} + \frac{b}{2} + \frac{c}{2} = \frac{a+b+c}{2}$ $= \frac{\left(-\frac{b}{a}\right)}{2}$ $= \frac{-(-6)}{2}$ $= 3$	1	

Question	Suggested solution	Marking guidelines	Markers Notes
(ii)	$\frac{2}{a} + \frac{2}{b} + \frac{2}{c} = \frac{2(\beta\gamma + \alpha\gamma + \alpha\beta)}{4\beta\gamma}$ $= \frac{2\left(\frac{c}{a}\right)}{\left(-\frac{a}{a}\right)}$ $= \frac{2(4)}{(-2)}$ $= -4$	1	
f)	$\int \frac{3x+5}{\sqrt{x+2}} = \int \frac{3(u-2)+5}{\sqrt{u}} du$ $u = x+2 \quad = \int u^{\frac{1}{2}} (3u-1) du$ $x = u-2$ $\frac{du}{dx} = 1 \quad = \int 3u^{\frac{1}{2}} - u^{-\frac{1}{2}} du$ $\therefore du = dx = \frac{3u^{\frac{3}{2}}}{(\frac{3}{2})} - \frac{u^{\frac{1}{2}}}{(\frac{1}{2})} + C$ $= 2\sqrt{u^3} - 2\sqrt{u} + C$ $= 2\sqrt{(x+2)^3} - 2\sqrt{x+2} + C$	1 1 1	$\frac{1}{\sqrt{u}} = u^{-\frac{1}{2}}$ not $u^{\frac{1}{2}}$ * these are not like terms $\therefore \neq 2u^{\frac{1}{2}}$ * Answer must be in same form as question. i.e surds not fractional indices and in terms of x * Mark still awarded if left as fractional indice

Question	Suggested solution	Marking guidelines	Markers Notes
12.(a)	$u = e^x \quad v = \tan^{-1} x$		
	$u' = e^x \quad v' = \frac{1}{1+x^2}$		
		1	
	$\frac{d}{dx} (e^x \cdot \tan^{-1} x) = \frac{e^x}{1+x^2} + \tan^{-1} x \cdot e^x$	1	
	$= e^x \left(\frac{1}{1+x^2} + \tan^{-1} x \right)$		
(b)	$LHS = \frac{1 + \cos x + \sin x}{1 - \cos x + \sin x}$		
	$= \left(1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2} \right) \div \left(1 - \frac{(1-t^2)}{1+t^2} + \frac{2t}{1+t^2} \right)$	1	
	$= \left(\frac{1+t^2 + 1-t^2 + 2t}{1+t^2} \right) \div \left(\frac{1+t^2 - 1+t^2 + 2t}{1+t^2} \right)$		
	$= \left(\frac{2+2t}{1+t^2} \right) \div \left(\frac{2t^2+2t}{1+t^2} \right)$		
	$= \frac{2(1+t)}{1+t^2} \times \frac{1+t^2}{2t(t+1)}$	1	
	$= \frac{1}{t}$		
	$= \frac{1}{\tan \frac{x}{2}}$		
	$= \cot \frac{x}{2}$	1	
	$= RHS$		

Question	Suggested solution	Marking guidelines	Markers Notes
(c)	$5^n - 4n - 1$ divisible by 4 for $n \geq 2$ Step 1: Prove true for $n=2$ $5^2 - 4(2) - 1 = 25 - 8 - 1$ $= 16$ ($\div$ by 4) $\therefore$ True for $n=2$ Step 2: Assume true for $n=k$ i.e. $5^k - 4k - 1 = 4M$ where M is $5^k = 4M + 4k + 1$ a positive integer Step 3: Prove true for $n=k+1$ i.e. $5^{k+1} - 4(k+1) - 1$ is divisible by 4 $5^{k+1} - 4(k+1) - 1 = 5 \cdot 5^k - 4k - 4 - 1$ $= 5(4M + 4k + 1) - 4k - 5$ (from step 2) $= 20M + 20k + 5 - 4k - 5$ $= 20M + 16k$ $= 4(5M + 4k)$ ($\div$ by 4) 1 Step 4: Conclusion True for $n=k+1$ if true for $n=k$ True for $n=2$ $\therefore$ by induction, true for all integers $n \geq 2$	1.	

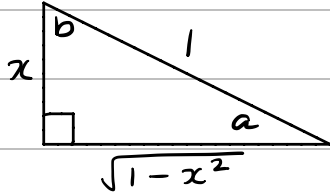
Question	Suggested solution	Marking guidelines	Markers Notes
(d) (i)	For $f(x) = 2\cos^{-1}(x-1)$, $1 \leq x \leq 2$. when $x = 1$ $f(1) = 2\cos^{-1}(1-1)$ $= 2\cos^{-1} 0$ $= 2\left(\frac{\pi}{2}\right)$ $= \pi$ Domain $\cos^{-1} x$ $-1 \leq x \leq 1$ $-1 \leq x-1 \leq 1$ $0 \leq x \leq 2$ $\therefore$ Domain $f(x)$: $1 \leq x \leq 2$ Range $\cos^{-1} x$ $0 \leq \cos^{-1}(x-1) \leq \pi$ $0 \leq 2\cos^{-1}(x-1) \leq 2\pi$ $\therefore 0 \leq y \leq 2\pi$ $\therefore$ Range $f(x)$: $0 \leq y \leq \pi$ 		1- correct shape and x-intercept at $x=2$ 1- coordinates of endpoint at $x=1$.
(ii)	$y = 2\cos^{-1}(x-1)$ for $0 \leq y \leq \pi$ $x = 2\cos^{-1}(y-1)$ $\frac{x}{2} = \cos^{-1}(y-1)$ $\cos\left(\frac{x}{2}\right) = y-1$ $\cos\left(\frac{x}{2}\right) + 1 = y$ $\therefore f^{-1}(x) = \cos\left(\frac{x}{2}\right) + 1$. $0 \leq x \leq \pi$ ① ①		

Question	Suggested solution	Marking guidelines	Markers Notes
13.(a)(i)	$f(x) = x^3 + 2x - 7$		
	$f(1) = 1^3 + 2(1) - 7$		
	$= -4$		
	< 0		
	$f(2) = 2^3 + 2(2) - 7$		
	$= 5$		
	> 0	1	
	Since $f(1) < 0$ and $f(2) > 0$		
	there is a root between		
	1 and 2.	1	
(ii)	$f'(x) = 3x^2 + 2$		
	> 0 for all real x		
	$\therefore f(x)$ is monotonic increasing		
	$\Rightarrow$ graph cannot cross x -axis		
	more than once		
	Hence α is the only real root	1	
(iii)	$f(1.5) = 1.5^3 + 2(1.5) - 7$		
	$= -0.625$		
	$f'(1.5) = 3(1.5)^2 + 2$		
	$= 8.75$	1	
	$\alpha_2 = \alpha_1 - \frac{f(\alpha_1)}{f'(\alpha_1)}$		
	$= 1.5 - \frac{(-0.625)}{8.75}$		
	≈ 1.6 (1dp)	1	

Question	Suggested solution	Marking guidelines	Markers Notes
(b)	$\angle ABD = \angle ACD$ ($\angle$'s in same segment) 1 $\angle ABF = \angle BFE$ (alternate $\angle$'s, $FE \parallel AB$) 1 $\therefore \angle ACD = \angle DFE$ $\therefore DCEF$ is a cyclic quadrilateral (exterior $\angle$ equal to opposite interior $\angle$) 1		
(c)	$\int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx = \left[\sin^{-1} \frac{x}{2} \right]_{\sqrt{2}}^{\sqrt{3}}$ $= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \frac{\sqrt{2}}{2}$ $= \frac{\pi}{3} - \frac{\pi}{4}$ $= \frac{4\pi - 3\pi}{12}$ $= \frac{\pi}{12}$	1 1 1 1	

[illegible]

Question	Suggested solution	Marking guidelines	Markers Notes
	$y = \frac{\left(t + \frac{2}{t}\right)^2 + t^2}{2}$ $= \frac{t^2 + 4 + \frac{4}{t^2} + t^2}{2}$ $= \frac{2t^2 + 4 + \frac{4}{t^2}}{2}$ $= \cancel{2} \left(t^2 + 2 + \frac{2}{t^2} \right)$ $= t^2 + 2 + \frac{2}{t^2} \quad \textcircled{2}$	1	
	sub ① into ② $y = \left(-\frac{2}{x}\right)^2 + 2 + \frac{2}{\left(-\frac{2}{x}\right)^2}$ $= \frac{4}{x^2} + 2 + \frac{2}{\left(\frac{4}{x^2}\right)}$ $= \frac{4}{x^2} + 2 + \frac{x^2}{2}$	1	

Question	Suggested solution	Marking guidelines	Markers Notes
14(a)	$\sin^{-1}x - \cos^{-1}x = \frac{\pi}{3}$ $\text{Let } a = \sin^{-1}x \quad b = \cos^{-1}x$ $\sin a = x \quad \cos b = x$  $a - b = \frac{\pi}{3} \quad \text{①}$ $a + b = \frac{\pi}{2} \quad \text{②}$ $\text{①} + \text{②} \quad 2a = \frac{\pi}{3} + \frac{\pi}{2}$ $2a = \frac{5\pi}{6}$ $a = \frac{5\pi}{12}$ $\therefore \sin^{-1}x = \frac{5\pi}{12}$	Alternatively: $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$ $\therefore \cos^{-1}x = \frac{\pi}{2} - \sin^{-1}x$ $\sin^{-1}x - \left(\frac{\pi}{2} - \sin^{-1}x\right) = \frac{\pi}{3}$ $2\sin^{-1}x - \frac{\pi}{2} = \frac{\pi}{3}$ $2\sin^{-1}x = \frac{5\pi}{6}$ $\sin^{-1}x = \frac{5\pi}{12}$	
(b) (i)	$\sin 2t - \sqrt{3}\cos 2t = R \sin(2t - \alpha)$ $R = \sqrt{1^2 + (\sqrt{3})^2} \quad \tan \alpha = \frac{\sqrt{3}}{1}$ $= \sqrt{1+3} \quad \alpha = \frac{\pi}{3}$ $= \sqrt{4}$ $= 2$ $\therefore \sin 2t - \sqrt{3}\cos 2t = 2 \sin\left(2t - \frac{\pi}{3}\right)$	1 1 1	
(ii)	$x = 2 \sin\left(2t - \frac{\pi}{3}\right)$ $\dot{x} = 4 \cos\left(2t - \frac{\pi}{3}\right)$ $\ddot{x} = -8 \sin\left(2t - \frac{\pi}{3}\right)$ $= -4 \left(2 \sin\left(2t - \frac{\pi}{3}\right)\right)$ $= -4x$ $= -2^2 x$	1 1	

$\therefore$ since acceleration is in the form $-n^2x$, the particle is moving in SHM

Question	Suggested solution	Marking guidelines	Markers Notes
(iii)	$n = 2$	1	
	$T = \frac{2\pi}{n}$		
	$= \frac{2\pi}{2}$		
	$= \pi$		
(iv)	$2 \sin(2t - \frac{\pi}{3}) = 2$	1	
	$\sin(2t - \frac{\pi}{3}) = 1$		
	$2t - \frac{\pi}{3} = \frac{\pi}{2}$		
	$2t = \frac{\pi}{2} + \frac{\pi}{3}$		
	$= \frac{5\pi}{6}$		
	$t = \frac{5\pi}{12}$		
(c) (i)	$\therefore$ Particle first reached 2 metres right of the origin at $\frac{5\pi}{12}$ seconds	1	
	$y = 60t \sin \theta - 5t^2$ ①	1	
	$\frac{dy}{dt} = 60 \sin \theta - 10t$		
	Maximum height when $y' = 0$		
	$0 = 60 \sin \theta - 10t$		
	$10t = 60 \sin \theta$		
	$t = 6 \sin \theta$ ②		
	sub ② into ①		
	$y = 60(6 \sin \theta) \sin \theta - 5(6 \sin \theta)^2$		
	$= 360 \sin^2 \theta - 180 \sin^2 \theta$		
	$= 180 \sin^2 \theta$ as required	1	

Question	Suggested solution	Marking guidelines	Markers Notes
14(c)(ii)	$x = 60 + \cos \theta$ ③ sub ② into ③ $x = 60(6 \sin \theta) \cos \theta$ $= 360 \sin \theta \cos \theta$ $= 180(2 \sin \theta \cos \theta)$ $= 180 \sin 2\theta$ as required	1	
(iii)	For best viewing $90 < x < 115$ $90 < 180 \sin 2\theta < 115$ $\frac{1}{2} < \sin 2\theta < \frac{23}{36}$ $30^\circ < 2\theta < 40^\circ$ or $140^\circ < 2\theta < 150^\circ$ $15^\circ < \theta < 20^\circ$ $70^\circ < \theta < 75^\circ$	1	
	Also $y \geq 140$ $180 \sin^2 \theta \geq 140$ $\sin^2 \theta \geq \frac{7}{9}$ $\theta \geq \sin^{-1}\left(\frac{\sqrt{7}}{3}\right)$ $\geq 62^\circ$	1	
	$\therefore 70^\circ < \theta < 75^\circ$	1	